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ADBMS-Expt 3

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Experiment 03

Easy

(A):

You are given with the Employee schema; the task is to find the highest score for each respective department.

For eg: department D1 have the highest score as 5.28. Hence in the output, final updated score for D1 should be 5.28.

Same with the other departments i.e., D2, D3 and D4.

EID	DEPT	SCORES
1	D1	1
2	D1	5.28
3	D1	4
4	D2	8
5	D1	2.5
6	D2	7
7	D3	9
8	D4	10.2

Employee/postgres@PostgreSQL 18

No limit

Query Query History

5

DEPT VARCHAR(10),

6

SCORES DECIMAL(5,2)

7

);

8

9

INSERT INTO Employee (EID, DEPT, SCORES)

10

VALUES

11

(1, 'D1', 1.00),

12

(2, 'D1', 5.28),

13

(3, 'D1', 4.00),

14

(4, 'D2', 8.00),

15

(5, 'D1', 2.50),

16

(6, 'D2', 7.00),

17

(7, 'D3', 9.00),

18

(8, 'D4', 10.20);

19

20

-- 1.

21

update Employee e

22

set SCORES=(select max(SCORES) from Employee where DEPT=e.DEPT);

23

24

Select * from Employee;

25

Data Output Messages Notifications

SQL

Showing rows: 1 to 8

	eid [PK] integer	dept character varying (10)	scores numeric (5,2)
1	1	D1	5.28
2	2	D1	5.28
3	3	D1	5.28
4	4	D2	8.00
5	5	D1	5.28
6	6	D2	8.00
7	7	D3	9.00
8	8	D4	10.20



Experiment 03

Medium

(C): SALES ANALYSIS PART 1

A company maintains records of products sold by different salespersons. The management wants to prepare a sales performance report by identifying the salesperson(s) who generated the highest overall sales revenue. If multiple salespersons have the same highest total sales amount, all of them must be included in the final report. Return the seller ID(s) satisfying this condition.

Employee/postgres@PostgreSQL 18

No limit

Query Query History

63

(4, 'Pam', 'Boston', 15);

64

65

INSERT INTO Company

66

VALUES

67

(1, 'RED', 'Boston'),

68

(2, 'ORANGE', 'New York'),

69

(3, 'YELLOW', 'Boston'),

70

(4, 'GREEN', 'Austin');

71

72

INSERT INTO Orders

73

VALUES

74

(1, '2024-01-10', 1, 1, 1200),

75

(2, '2024-01-12', 2, 1, 800),

76

(3, '2024-01-15', 3, 2, 2500),

77

(4, '2024-01-18', 1, 3, 1500),

78

(5, '2024-01-22', 4, 2, 700),

79

(6, '2024-01-25', 2, 3, 2000),

80

(7, '2024-01-28', 3, 4, 3000),

81

(8, '2024-01-30', 4, 4, 200);

82

83

84

Select seller_id from

85

Orders

86

group by seller_id

87

having sum(amount) in (Select max(max_amt) from

88

(Select seller_id, sum(amount) as max_amt from Orders group by seller_id

89

))

Data Output Messages Notifications

SQL

Showing rows: 1 to 1

Page

	seller_id integer
1	3



Experiment 03

Hard

(B):

A company maintains employee salary records for different departments. The management wants to prepare a report containing employees whose salaries fall among the **top three highest distinct salary values** within their respective departments. If multiple employees receive the same qualifying salary, all such employees must be included in the final report. Display the department name, employee name, and salary for every qualifying employee.

Query

Query History

```

122  (1, 'Joe', 85000, 1),
123  (2, 'Henry', 80000, 2),
124  (3, 'Sam', 60000, 2),
125  (4, 'Max', 90000, 1),
126  (5, 'Janet', 69000, 1),
127  (6, 'Randy', 85000, 1),
128  (7, 'Will', 70000, 1);
129
130  select
131  d.name as Department,
132  e.name as Employee,
133  e.salary as salary
134  from Employee e
135  join
136  Department d
137  on e.departmentId=d.id
138  where
139  (
140  select count(distinct e1.salary) from Employee e1
141  where e.departmentId=e1.departmentId and e.salary<e1.salary
142  ) <3
143  order by Department, Employee, salary desc
144
    
```

Data Output

Messages

Notifications

	department character varying (50)	employee character varying (50)	salary integer
1	IT	Joe	85000
2	IT	Max	90000
3	IT	Randy	85000
4	IT	Will	70000
5	Sales	Henry	80000
6	Sales	Sam	60000